Jeff Santos

AI Engineer

+351 911 012 161 | Lisbon, Portugal (Remote)

jmsapps@outlook.com | jmsapps.ca | linkedin.com/in/jmsapps | github.com/jmsapps

Summary

Full-stack engineer specializing in AI-enabled data platforms and cloud-native systems. Experienced with Python, TypeScript, React, Svelte, NestJS/Node.js, and PostgreSQL/GraphQL. Builds reliable ingestion pipelines for Retrieval-Augmented Generation (RAG) using Azure Document Intelligence, validation/observability best practices, and CI/CD automation across Azure and GCP.

Skills

Frontend: CSS, DOM, HTML, JavaScript, Liquid, React Native, React.js, Redux, Svelte, Styled Components, Swift, TypeScript
Backend: ExpressJS, NestJS, Nim, Node.js, Python, Tornado, TypeScript, Django
Data & AI: RAG, Azure Document Intelligence, Hugging Face (transformations), Data validation, Observability, Reporting
Database: GraphQL, Hasura, MongoDB, MySQL, PostgreSQL, SQL, NoSQL, OData, ORM
Cloud & DevOps: Azure, Google Cloud (Cloud Run, Cloud Scheduler), AWS (EC2, S3), Docker, Terraform, GitHub Actions, Linux, Nginx, REST, SOAP

Experience

AI Engineer (Data Ingestion & RAG) August 2025 – Present
Tactable *Toronto, ON*

- Owned the data ingestion pipeline for a Retrieval-Augmented Generation (RAG) system for a large enterprise client (NDA), driving delivery from proposal through implementation by partnering with stakeholders and product leadership.
- Reduced end-to-end ingestion time from 15 minutes to as low as 20 seconds (**97.8% faster**; 45× speedup), improving iteration cycles and system freshness.
- Built ingestion and parsing workflows using **Azure** and **Azure Document Intelligence**, with custom validation, quality checks, and automated reporting for downstream consumers.
- Implemented observability best practices (structured logs/metrics, traceable runs, failure summaries) to improve reliability and speed up incident resolution.
- Integrated supplemental **Hugging Face** models for domain-specific data transformations and normalization where deterministic rules were insufficient.

Software Engineer (Compliance & Integrations) January 2025 – August 2025
Tactable *Toronto, ON*

- Led system design and development for a compliance product, translating requirements into an implementation roadmap for the development team.
- Built and maintained SOAP services (Yardi Voyager 7S, MSI) and managed a Hasura/NestJS backend with custom PostgreSQL queries.
- Implemented CI/CD pipelines with GitHub Actions and provisioned Cloud Run services using Terraform on GCP.
- Created local testing tools for Cloud Scheduler endpoints and maintained test coverage with Jest for reliable deployments.
- Drove Python/Django performance work: reduced a critical endpoint from 16s to 0.5s (32× faster; -96.9% latency).
- Untangled and modularized a 3,000+ line Python ingestion script; added 100% test coverage to stabilize onboarding and changes.
- Built a custom Python-based MongoDB migration version control system enabling repeatable environment migrations and safer deployments.

Lead Software Engineer, Full Stack June 2023 – January 2025
Switch Health *Toronto, ON*

- Delivered major performance improvements on Azure cloud solution, including **200% faster** API response times in critical areas.
- Redesigned relational database structure while preserving data integrity for **4M+ users**.

- Cut feature implementation time from **two weeks to one day** by building a reusable pipeline for new test types and internal tooling.
- Enabled B2B integrations through a White Label platform, **doubling profitability**.
- Increased conversion rates by **20%** with targeted UX enhancements (Liquid/Vanilla JS) and iterative experimentation.
- Mentored junior developers and conducted code reviews across the product lifecycle.

Software Engineer II, Full Stack — January 2023 – June 2023
Switch Health — *Toronto, ON*

- Transformed the platform into a White Label solution, enabling B2B integrations and improving business scalability.
- Migrated state management to Redux Toolkit, reducing bundle size by 60% and development time by 25% while improving maintainability.
- Led a UI redesign from Figma to production, collaborating closely with Product and QA to ship quality releases.
- Refactored the front-end application into modular, reusable components to accelerate delivery.

Software Engineer I, Full Stack — April 2022 – December 2022
Switch Health — *Toronto, ON*

- Restructured the project into a well-organized folder hierarchy, accelerating development by 20%.
- Implemented best practices and reduced technical debt, improving team collaboration and code maintainability.
- Designed and developed RESTful APIs for complex test types, optimizing back-end integration for high-throughput data.
- Diagnosed and resolved critical software issues, enhancing system stability and performance.

Projects

Cloud Storage Solution | *Family Albums*

- Built a scalable cloud storage platform with a React/TypeScript front-end and a Python/Tornado back-end, integrated with AWS EC2, S3, and PostgreSQL.
- Implemented Stripe for secure payment processing, leveraging SSL certificates and encryption best practices.
- Automated operations using Slack webhooks and Docker, improving monitoring and deployment workflows.

Content Management System for Realtors | *Realty Host*

- Developed a CMS for Realtors, enabling tenant-specific UI customization while maintaining a unified, scalable back-end architecture.
- Integrated TRREB's Ampre API via OData, optimizing ingestion and streamlining property search.
- Implemented OAuth 2.0 and JWT authentication for secure, scalable user management.

iOS Weather Application | *Daily Forecast*

- Developed an iOS weather app using Swift and Core Data, processing real-time data from the OpenWeather API.
- Achieved 20K+ downloads with 1K+ active daily users by iterating on UI/UX and retention features.
- Monetized the app via Google AdMob with strategic ad placements.

Front-End Web Framework | *NTML*

- Developed a lightweight, React-inspired front-end framework using Nim for component-based UI development.
- Optimized workflows with Nim Macros/Templates to reduce boilerplate and improve maintainability.
- Designed the framework to be extensible and easy to integrate with existing web architectures.

Education

Seneca College, Computer Programming — North York, ON